

Kefan Xu

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Summary

I am a Ph.D. Student in Human-Centered Computing at the Georgia Institute of Technology, operating at the intersection of HCI, CSCW, and Health Informatics. Rooted in a commitment to health equity, my research develops human-centered systems as probes to facilitate sensemaking, coping, and dynamic management strategies for chronic conditions. Utilizing an “ecological informatics” approach, I design and develop functional mobile applications evaluated through longitudinal deployment studies. Looking forward, I aim to explore how Human-Centered AI can further enhance situated care ecologies. As an educator, I embrace a multidisciplinary, student-centered teaching philosophy that bridges theoretical HCI concepts with practical software development, empowering mentees to tackle real-world health challenges and succeed in both academia and industry.

Education

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| 2022 – Present | Georgia Institute of Technology – Atlanta, GA
Ph.D. in Human-Centered Computing
Advisor: Dr. Rosa I. Arriaga. <i>GPA: 4.0/4.0.</i> |
| 2019 – 2021 | University of Michigan – Ann Arbor, MI
M.S. in Information Science
Advisor: Dr. Mark W. Newman. <i>GPA: 4.0/4.0.</i> |
| 2015 – 2019 | New York University – New York City, NY & Shanghai, China
B.S. in Data Science B.S. in Interactive Media Arts (Doubled)
<i>GPA: 3.6/4.0.</i> |

Honors and scholarships

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| 2025 | OMSCS Pre-Doc Fellowship (Georgia Institute of Technology: 3 Semesters) |
| 2022 | Google Health Equity Research Initiative (Google) |
| 2021 | UMSI Master Thesis (MTOP) Grant (University of Michigan) |
| 2021 | IHPI/HSR Summer Student Fellowship (University of Michigan) |
| 2019 | NYU Founders Day Award (New York University) |
| 2016 | Tel Aviv University Summer Scholar Program (Tel Aviv University) |

Publications

Peer-reviewed Full Conference Papers

- C6 2026 **Human-centered Perspectives on a Clinical Decision Support System for Intensive Outpatient Veteran PTSD Care**
Cynthia M. Baseman, Myeonghan Ryu, Nathaniel Swinger, **Kefan Xu**, Andrew M. Sherrill, Rosa I. Arriaga.
Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26).
- C5 2025 **Explainable AI for Daily Scenarios from End-Users' Perspective: Non-Use, Concerns, and Ideal Design**
Lingqing Wang, Chidimma Lois Anyi, **Kefan Xu**, Yifan Liu, Rosa I. Arriaga, Ashok K. Goel.
Proceedings of the 2025 ACM Designing Interactive Systems Conference (DIS '25).
- C4 2025 **Understanding the Temporality of Informal Caregivers' Sense-Making on Conflicts and Life-Changing Events through Online Health Communities**
Kefan Xu, Cynthia M. Baseman, Nathaniel Swinger, Myeonghan Ryu, Rosa I. Arriaga.
Proc. ACM Hum.-Comput. Interact. (CSCW '25).
- C3 2024 **Understanding the Effect of Reflective Iteration on Individuals' Physical Activity Planning**
Kefan Xu, Xinghui Yan, Myeonghan Ryu, Mark W. Newman, Rosa I. Arriaga.
Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24).
- C2 2024 **Using Sensor-Captured Patient-Generated Data to Support Clinical Decisionmaking in PTSD Therapy**
Hayley I. Evans, Myeonghan Ryu, Theresa Hsieh, Jiawei Zhou, **Kefan Xu**, Kenneth W. Akers, Andrew M. Sherrill, Rosa I. Arriaga.
Proc. ACM Hum.-Comput. Interact. (CSCW '24).
- C1 2022 **Understanding People's Experience for Physical Activity Planning and Exploring the Impact of Historical Records on Plan Creation and Execution**
Kefan Xu, Xinghui Yan, Mark W. Newman.
CHI Conference on Human Factors in Computing Systems (CHI '22).

Workshop Papers

- W1 2025 **Facilitating Individuals' Sensemaking about Sedentary Behavior via Contextualized Data**
Kefan Xu, Rosa I. Arriaga.
Workshop at the ACM CHI Conference on Human Factors in Computing Systems (CHI '25).

- 2026 **Postphenomenological Insights from an In-the-Wild Collective Sensing System Deployment for PTSD Therapy**
Swinger, N.J., Ryu, M., Baseman, C.M., **Xu, K.**, Sherrill, A.W., Arriaga, R.I.
Accepted, Proc. ACM Hum.-Comput. Interact. (CSCW '26).
- 2026 **“I’m a person—not a patient or diagnosis”: Centering the Person Living with Dementia’s Perspective on Caregiver Dynamics and Technology Design**
Yuan, H., **Xu, K.**, Arriaga, R.I., Hub, J.
Accepted, ACM Transactions on Computing for Healthcare.
- Towards Ecological Informatics: Navigating the Evolving Landscape for Chronic Disease Management**
Xu, K., Thukral, A., Soneji, R., Ghahremani, S., Fayfman, M., Schechter, C.M., Arriaga, R.I.
In submission.
- Beyond the Prompt Box: Co-designing Multimodal Generative AI with University Students in Everyday Contexts**
Zhang, C., Huang, J., **Xu, K.**, Magerko, B.
In submission.

Research Projects

- 2025 – Present **Trackya**
Investigated how contextual data influences sense-making within personal informatics. Designed the mobile application’s UI dashboard and developed comprehensive, three-phase study protocols. Presented early findings at a CHI 2026 workshop.
- 2025 – Present **Moodloop**
Investigated individuals’ sense-making on subjective mental reporting. Designed the personal informatics interface, developed comprehensive interview protocols, and deployed the application to evaluate longitudinal user experiences.
- 2022 – Present **Diabetes Ubiquitous Computational Sensing System (DUCSS)**
Innovated self-management platforms for diabetes patients. Designed system architectures ensuring that patients and caregivers interact with the system individually, rather than as dyads, and screened participants via comprehensive eligibility surveys.
- 2022 – 2025 **Prolonged Exposure Collective Sensing System (PECSS)**
Evaluated and implemented clinical tools for PTSD management alongside Emory University clinical partners.

2021 – 2023 **Physical Activity Planning Systems (Planneregy & Physicify)**
Developed and deployed iOS applications to study how historical records shape exercise habits. Orchestrated 28-day and 42-day user studies to evaluate the reflection-iteration framework.

Research experience

2022 – Present **Graduate Research Assistant, Ubicomp Health and Wellness Lab**
Georgia Institute of Technology
Innovated self-management platforms for diabetes patients (DUCSS). Designed system architectures ensuring that patients and caregivers interact with the system individually, rather than as dyads, and screened participants via comprehensive eligibility surveys. Evaluated clinical tools for the Prolonged Exposure Collective Sensing System (PECSS) and analyzed chronic disease family caregiving dynamics via Reddit.

2019 – 2022 **Graduate Research Assistant, School of Information**
University of Michigan
Developed and deployed iOS applications (Planneregy, Physicify) to study how historical records shape exercise habits. Orchestrated 28-day and 42-day user studies. Implemented the "Pace to Plan" platform using React Native to develop low-burden hybrid Experience Sampling Methods (ESM) for older adults managing chronic fatigue.

2021 – 2022 **Research Intern, Michigan Medicine**
University of Michigan
Designed and deployed Taperology, a clinician-facing web platform featuring a custom taper calculator algorithm, empowering clinicians to safely formulate tailored schedules for benzodiazepine misuse therapy.

Teaching experience

Fall 2025 –
Summer 2026 **Instructor, CS8001ODW: Developing Mobile Experience for Well-Being**
Georgia Institute of Technology
Designed and lectured a 16-week course covering the theoretical and developmental knowledge required for health informatics and mobile app design.

Spring –
Summer 2024 **Graduate Teaching Assistant, CS 7470: Mobile and Ubiquitous Computing**
Georgia Institute of Technology
Mediated in-class activities and led assignments under Prof. Thomas Ploetz and Prof. Clint Zeagler.

Presentations and invited talks

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| 2026 | Paper Presentation
<i>CSCW Conference, Bergen, Norway</i> |
| 2026 | Trackya Project Presentation
<i>CHI 2026 Workshop</i> |
| 2024 | Research Showcase
<i>IPaT Research Showcase</i> |
| 2023 | Poster Presentation
<i>CRIDC Poster Competition</i> |

Mentorship and service

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| 2022 – Present | Student Mentorship
Supervised and mentored Cynthia Baseman (PhD Student, GaTech), Diptark Bose (Master's Student, now SDE at Apple), Hao Yuan (Master's Student, now PhD at Stevens Institute), and Abhinav Thukral (Master's Student, now UX Designer at Lucidchart) on system development and study protocols. |
| 2022 – Present | Conference Reviewer
Actively reviewed academic papers for ACM CHI (2022, 2025, 2026), ACM IMWUT (2023, 2025), IEEE ICHI (2022), and CHI Late-Breaking Work (2022). |

Technical skills

Programming languages

React Native, Python, Java, C++, C#, Swift, HTML/CSS/JavaScript, R, MySQL

Software & Tools

Figma, Expo, Git, Unity, Overleaf, Adobe Creative Suite, Tableau

Research Methods

Contextual Inquiry, Affinity Mapping, Ethnographic Research, Thematic Analysis, Usability Evaluation, Prototyping, Co-Design